



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

**SECD 2613**  
**SYSTEM ANALYSIS AND DESIGN**

**PROJECT : PHASE 1**  
**Project Proposal**

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# TABLE OF CONTENTS

|                                                            |           |
|------------------------------------------------------------|-----------|
| <b>1.0 INTRODUCTION.....</b>                               | <b>2</b>  |
| <b>2.0 BACKGROUND STUDY.....</b>                           | <b>3</b>  |
| <b>3.0 PROBLEM STATEMENT.....</b>                          | <b>5</b>  |
| <b>4.0 PROPOSED SOLUTION.....</b>                          | <b>6</b>  |
| 4.1 Feasible Study.....                                    | 6         |
| 4.1.1 Technical Feasibility.....                           | 6         |
| 4.1.2 Operational Feasibility.....                         | 7         |
| 4.1.3 Economic Feasibility.....                            | 7         |
| <b>5.0 OBJECTIVES.....</b>                                 | <b>9</b>  |
| <b>6.0 SCOPE OF PROJECT.....</b>                           | <b>10</b> |
| <b>7.0 PROJECT PLANNING.....</b>                           | <b>11</b> |
| 7.1 HUMAN RESOURCE.....                                    | 11        |
| 7.2 WORK BREAKDOWN STRUCTURE (WBS).....                    | 12        |
| 7.3 PERT CHART.....                                        | 14        |
| 7.4 GANTT CHART.....                                       | 14        |
| <b>8.0 BENEFIT OVERALL SUMMARY OF PROPOSED SYSTEM.....</b> | <b>15</b> |

## 1.0 INTRODUCTION

As we know postgraduate students are often faced with different challenges of managing multiple tasks, from conducting research to drafting and publishing papers. Some postgraduate students even are part time students that need to manage their time to do their work and their studying for graduation. To produce a good paper writing and publication is not only based on what they have studied but also needs a complex process that demands good planning, organization, and time management. As system analysis and design students, we realize the need for an efficient and user-friendly system to the unique needs of postgraduate students engaged in academic research and publication.

The importance of proper task management in academic paper writing and publication cannot be overstated. Poorly managed tasks can lead to delays, decreased productivity, and, ultimately, suboptimal research results. Furthermore, the pressure to publish in reputable journals within a specific time frame makes the postgraduate journey more challenging.

Given these challenges, the development of a specialized task management system aimed at facilitating and streamlining the paper writing and publication process for postgraduates is both timely and essential. This proposed system aims to integrate various aspects of the research and publication process, giving postgraduate students a cohesive and organized platform to manage their tasks effectively.

This proposal describes our intention to design and develop a comprehensive Task Management System (TMS) that is specifically tailored to the needs of postgraduate students conducting academic research and publication. The proposed TMS will include features such as task listing, task tracking, deadline management, collaboration and communication tools, and resource allocation, reminders and alerts, among others, to improve overall efficiency and productivity in the paper writing and publication process.

Through this initiative, we hope to contribute to the academic community by providing postgraduate students with the tools and resources they need to navigate the complexities of academic paper writing and publication effectively. Our ultimate goal is to help postgraduate students have a more organized, more effective and less stressful research and publication journey, creating a conducive environment to academic excellence and innovation.

## 2.0 BACKGROUND STUDY

Paper writing and publications are one of the important components of postgraduate studies, serving as a vital platform for knowledge transfer, scholar recognition and career advancement. Thus, postgraduate students are expected to produce high quality paper writing and publications as evidence of their academic abilities and contributions to their respective fields of study. The process of writing and publishing papers is not only important academically, but it also plays a crucial role which impacts on postgraduate students' career paths.

There are a lot of challenges for postgraduate students when producing high quality paper writing and publications. Firstly, time management is one of the challenges for postgraduate students to balance multiple tasks at the same time. Besides, to keep track of various tasks, deadlines and resources required for paper writing and publication we need a good organization and planning skills. Moreover, collaborating with co-authors, advisors, and peers that work remotely or across different time zones and maintaining effective communication throughout the research and publication process can be challenging. Efficiently allocating and managing resources, such as research materials, data sets, and writing tools, is also crucial for the successful completion of academic papers and publications.

To address these challenges, we propose a Task Management System (TMS) with specialized functions. The TMS will include features such as task listing to create and manage a list of tasks associated with the research and publication process, allowing postgraduate students to break down the overall project into smaller, manageable tasks. Other than that, a timeline-based scheduling feature is added to help postgraduate students visualize and plan their research and publication activities over time, ensuring that all tasks and deadlines are effectively managed and met. Task Management System (TMS) will also integrate with Excel or similar spreadsheet software to facilitate data management, analysis, and visualization, streamlining the research process and enhancing data accuracy and consistency. Additionally, a progress tracking feature within the TMS will monitor the status of each task and the overall progress of the research and publication project, providing postgraduate students with real-time insights into their productivity and performance. Specialized collaboration and communication tools, also needed for fostering effective collaboration and enhancing the quality of academic papers and publications. Lastly, a reminders and alerts feature will notify postgraduate students of upcoming deadlines, overdue tasks, and important milestones, ensuring that they stay on track and manage their time effectively throughout the research and publication process.

Based on the challenges mentioned above, the proposed Task Management System (TMS) provides the necessary specialized functions to address the specific requirements of postgraduate students engaged in academic research and publication. The proposed TMS aims to combine the specialized functions mentioned above to provide a comprehensive and

user-friendly platform that improves the overall efficiency and productivity of the paper writing and publication process for postgraduate students.

### 3.0 PROBLEM STATEMENT

1. Fragmented Task Management:

Postgraduates struggle with fragmented task management processes, such as manually organizing research materials, coordinating with co-authors, and tracking deadlines for paper submission.

2. Communication Gaps:

Communication breakdowns between students and supervisors, as well as among co-authors, result in misunderstandings, missed deadlines, and delays in the publication process.

3. Resource Allocation Issues:

Difficulty in allocating resources effectively, such as time, research materials, and collaboration tools, leads to inefficiencies and suboptimal outcomes in research and publication endeavors.

4. Deadline Management:

Poor management of deadlines for literature review, data collection, analysis, and writing stages hampers progress and increases the likelihood of rushed, incomplete submissions.

5. Collaboration Challenges:

Coordinating tasks and sharing progress among co-authors or research team members becomes cumbersome due to reliance on disparate communication channels and tools.

6. Publication Requirement Compliance:

Lack of awareness or reminders regarding publication requirements, formatting guidelines, and submission procedures results in unnecessary revisions and rejections from journals or conferences.

## 4.0 PROPOSED SOLUTION

### 4.1 Feasible Study

Task Management System (TMS) is a system that allows postgraduate students to manage their assignments and projects efficiently. The system can be installed and applied on mobile devices such as mobile phones and tablets in order to enhance the functions of the Task Management System (TMS).

The Task Management System's (TMS) first function assists users or postgraduate students in allocating assignments so they can schedule a time to finish the projects. Apart from that, users can simply adhere to the timelines while working on projects because the system automatically saves the work timelines into a spreadsheet file each time a user opens and closes a project file. Additionally, the users are able to monitor the status of each work and the projects' progress. The users are required to enter the task progression from 0% to 100%. As a result, users can determine if the task is finished or only halfway done.

The second feature is the specialized collaboration and communication tools. The co-author of a project can be added by the author or postgraduate students who collaborate on the project. While the partners are now offline, users can still discuss projects with them by leaving a message in the communication space. As soon as they get the notification, the partners can reply to the message.

The third feature is the notification feature. The system will notify the users when they received messages, the upcoming due date, the existence of overdue tasks. The system helps users manage their time so they can complete their project publication and keeps the projects on schedule.

#### 4.1.1 Technical Feasibility

The Task Management System uses a mobile platform, it requires a mobile gadget and Internet access to use the system. A server and database system are required for this system since the system has to store user projects' progression and communicate with the team members. A security system guards the system to prevent confidential project information from being discovered by unidentified third parties.

#### 4.1.2 Operational Feasibility

The Task Management System requires permission to access dates, times, and files to track the progress of the project and manage the file. The message and project timeline are stored in a database within the system. Users can search the message history for messages sent by other team members. All the information and data are password-protected, the data can be accessed by the users with the password only.

#### 4.1.3 Economic Feasibility

##### Cost-Benefit Analysis

| Estimated Cost |                   |
|----------------|-------------------|
| Hardware       | RM 30,000         |
| Software       | RM 25,000         |
| Training       | RM 20,000         |
| Advertisement  | RM 2,500 per year |
| Maintenance    | RM 8,000 per year |
| User Support   | RM 3,000 per year |

| Assumption                       |     |
|----------------------------------|-----|
| Discount Rate                    | 10% |
| Sensitivity factor (cost)        | 1.1 |
| Sensitivity factor (benefits)    | 1.3 |
| Annual change in production cost | 5%  |
| Annual change in benefits        | 7%  |

| Estimated Benefits |           |
|--------------------|-----------|
| Saving             | RM 20,000 |
| Increase Sales     | RM 15,000 |

| Cost                                           | Year 0 | Year 1           | Year 2           | Year 3           | Year 4           | Year 5           |
|------------------------------------------------|--------|------------------|------------------|------------------|------------------|------------------|
| <b>Development costs</b>                       |        |                  |                  |                  |                  |                  |
| • Hardware                                     | 33,000 |                  |                  |                  |                  |                  |
| • Software                                     | 27,500 |                  |                  |                  |                  |                  |
| • Training                                     | 22,000 |                  |                  |                  |                  |                  |
| <b>Total</b>                                   | 82,500 |                  |                  |                  |                  |                  |
| <b>Production costs</b>                        |        |                  |                  |                  |                  |                  |
| • Advertisement                                |        | 2,750            | 2,888            | 3,032            | 3,184            | 3,343            |
| • Maintenance                                  |        | 8,800            | 9,240            | 9,702            | 10,187           | 10,696           |
| • User support                                 |        | 3,300            | 3,465            | 3,638            | 3,820            | 4,011            |
| <b>Annual production costs (Present value)</b> |        | 14,850<br>13,500 | 15,593<br>12,887 | 16,372<br>12,301 | 17,191<br>11,742 | 18,140<br>11,264 |
| <b>Accumulated costs</b>                       |        | 96,000           | 108,887          | 121,188          | 132,930          | 144,194          |

| Benefits                               | Year 0 | Year 1           | Year 2           | Year 3           | Year 4           | Year 5            |
|----------------------------------------|--------|------------------|------------------|------------------|------------------|-------------------|
| <b>Savings</b>                         |        | 26,000           | 33,800           | 43,940           | 57,122           | 74,259            |
| <b>Increase sales</b>                  |        | 19,500           | 25,350           | 32,955           | 42,842           | 55,695            |
| <b>Annual benefits (Present value)</b> |        | 45,500<br>41,364 | 59,150<br>48,884 | 76,895<br>57,772 | 99,964<br>68,277 | 129,954<br>80,691 |
| <b>Accumulated benefits</b>            |        | 41,364           | 90,248           | 148,020          | 216,297          | 296,988           |
| <b>Gain or Loss</b>                    |        | -54,636          | -18,639          | 26,832           | 83,367           | 152,974           |
| <b>Profitability Index</b>             | 1.85   |                  |                  |                  |                  |                   |

From the table shown above, the profitability index of the project is 1.85. Therefore, the project is a good investment due to the profitability index being higher than 1 even though there are some losses in the first year and second year with the total loss of RM 73,275. The expected profits for the fifth year and onwards are more than RM150,000. Consequently, this is a good project with a great return.

## 5.0 OBJECTIVES

- I. To establish a clear timeline for the research project, including milestones for literature review, data collection, analysis, and writing.
- II. To organize research materials efficiently, including articles, books, and data sources, for easy access during the writing process.
- III. To develop a structured outline for the paper, outlining key sections and subsections based on the research objectives and findings.
- IV. To allocate time for writing and revising drafts, ensuring adequate time for feedback and revisions from advisors or peers.
- V. To manage citations and references using a citation management tool, ensuring accuracy and consistency throughout the paper.
- VI. To coordinate with co-authors or collaborators, if applicable, to divide tasks and responsibilities effectively.
- VII. To track progress towards publication goals, including submission deadlines for conferences or journals.
- VIII. To implement strategies for dealing with challenges or setbacks, such as writer's block or unexpected data discrepancies.
- IX. To adhere to ethical guidelines and standards for research integrity throughout the writing and publication process.
- X. To prepare for the dissemination of research findings, including presentations at conferences or seminars, and promotion through social media or academic networks.

## 6.0 SCOPE OF PROJECT

### 6.1 TASK LISTING

- TASKS SEPARATION FOR PAPER WRITING AND PUBLICATION PROCESS

### 6.2 SCHEDULE

- TIMELINE FOR EACH TASK
- STUDENTS CAN ARRANGE TIME TO DO EACH TASK IN DIFFERENT PHASES

### 6.3 TRACK PROGRESS

- TRACKING THE TASKS MANUALLY
- REMINDER & ALERT
- MAKE SURE PROJECT FINISH ON TIME

### 6.4 COLLABORATION & COMMUNICATION

- GIVE FEEDBACK AND COMMENT
- IMPROVE EFFICIENCY OF PROJECT

## 7.0 PROJECT PLANNING

### 7.1 HUMAN RESOURCE



Project Manager: Tan Zhao Hong

- Manages tasks and progress of such tasks
- Assigns tasks to people

Designer: Chin Pei Wen

- Prepare rough design for website wireframe
- Refine design for website prototype

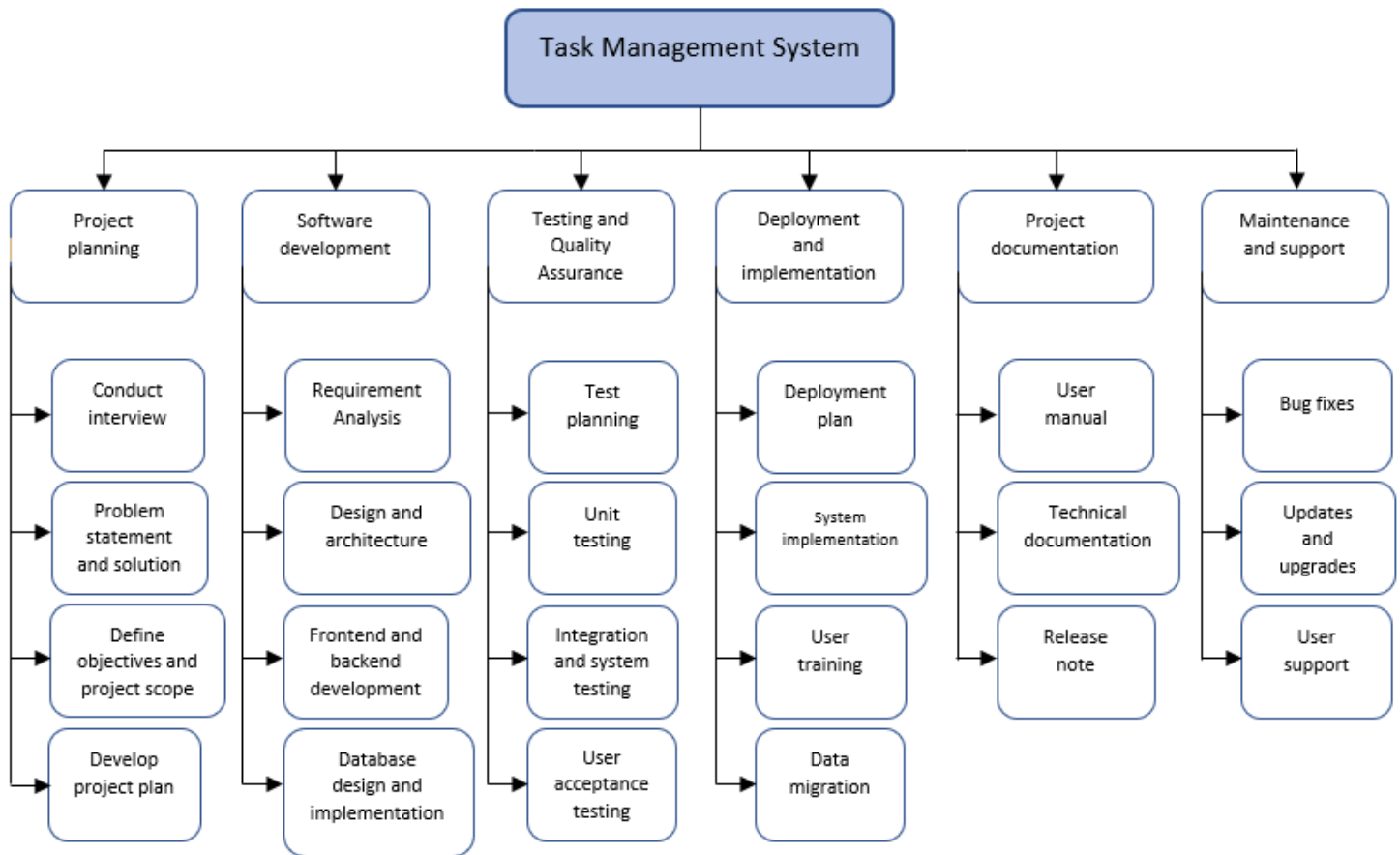
Programmer: Koo Xuan & Ling Yu Qian

- Build a programmable website based on design

Advisor: Chua Shang Yeet

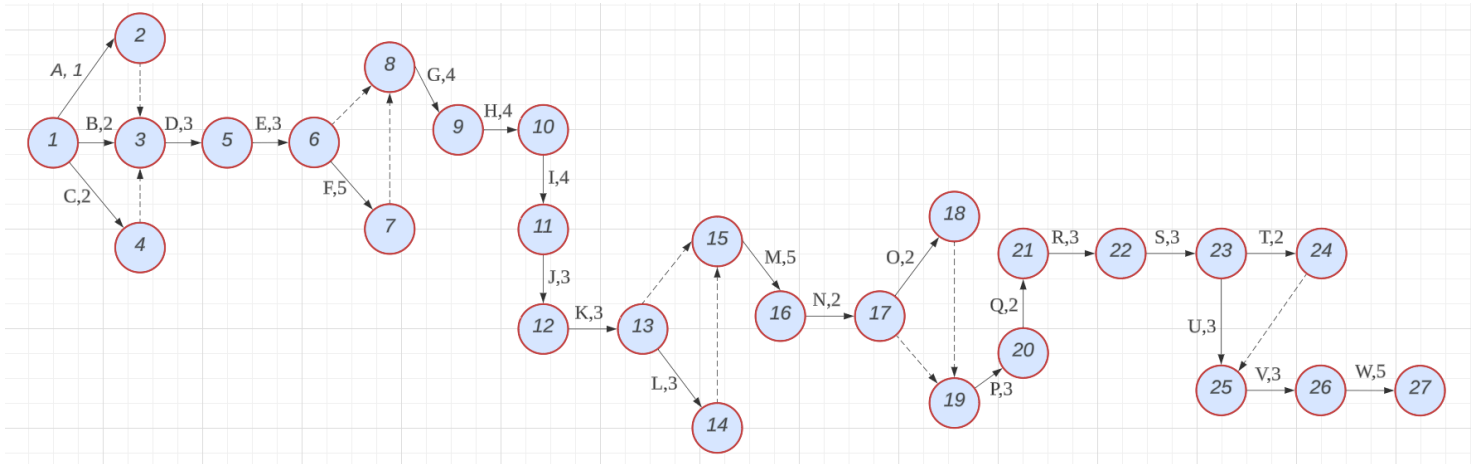
- Reviews progress
- Acknowledge and points out parts that can be improve

## 7.2 WORK BREAKDOWN STRUCTURE (WBS)

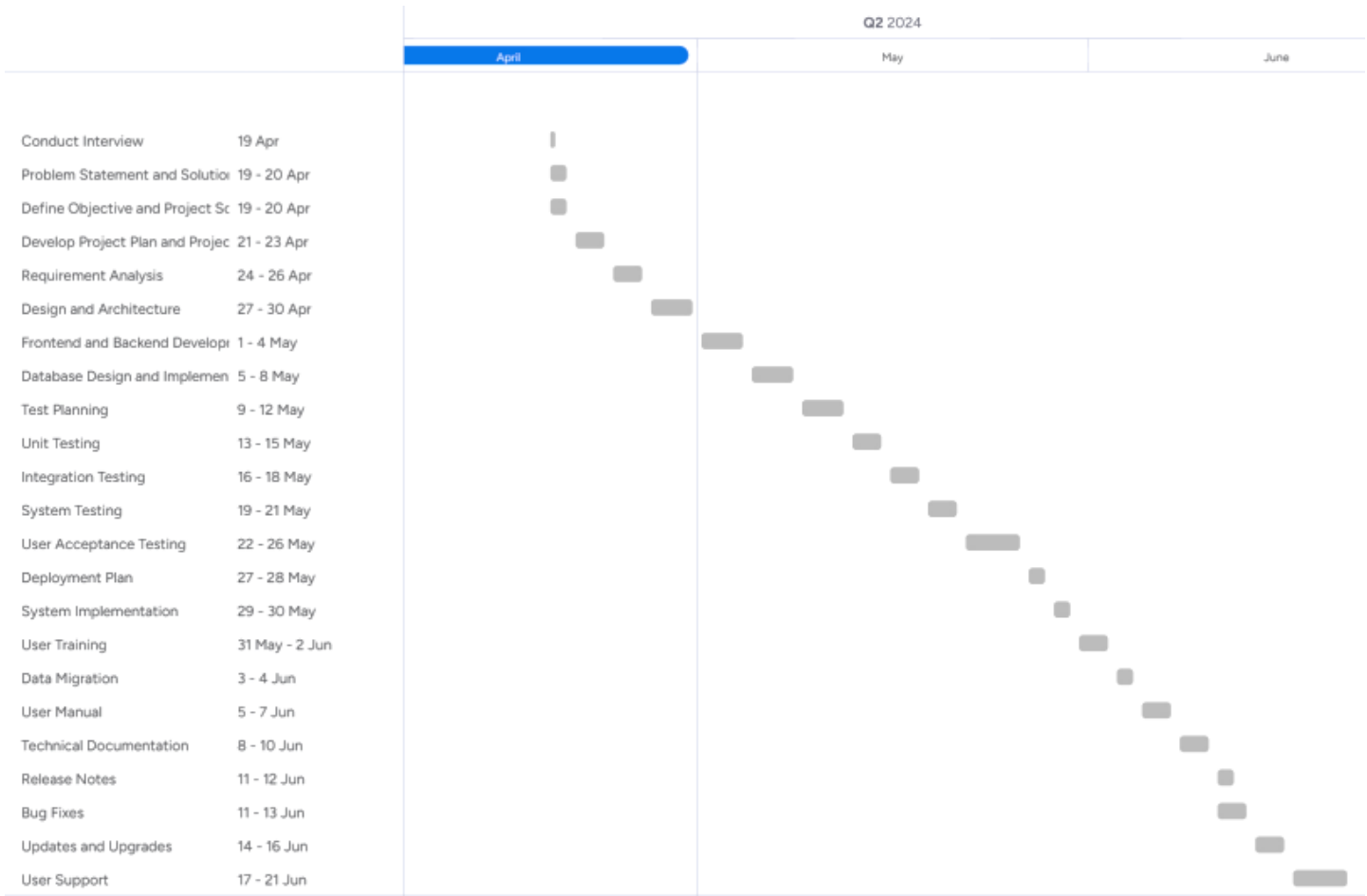


| Activity | Task                                  | Predecessor | Estimated Duration (Days) |
|----------|---------------------------------------|-------------|---------------------------|
| A        | Conduct Interview                     | None        | 1 (19/04/2024)            |
| B        | Problem Statement and Solution        | None        | 2 (19/04/2024-20/04/2024) |
| C        | Define Objective and Project Scope    | None        | 2 (19/04/2024-20/04/2024) |
| D        | Develop Project Plan and Project Flow | A,B,C       | 3 (21/04/2024-23/04/2024) |
| E        | Requirement Analysis                  | D           | 3 (24/04/2024-26/04/2024) |
| F        | Design and Architecture               | E           | 5 (26/04/2024-30/04/2024) |
| G        | Frontend and Backend Development      | E,F         | 4 (01/05/2024-04/05/2024) |
| H        | Database Design and Implementation    | G           | 4 (05/05/2024-08/05/2024) |
| I        | Test Planning                         | H           | 4 (09/05/2024-12/05/2024) |
| J        | Unit Testing                          | I           | 3 (13/05/2024-15/05/2024) |
| K        | Integration Testing                   | J           | 3 (16/05/2024-18/05/2024) |
| L        | System Testing                        | K           | 3 (19/05/2024-21/05/2024) |
| M        | User Acceptance Testing               | K,L         | 5 (22/05/2024-26/05/2024) |
| N        | Deployment Plan                       | M           | 2 (27/05/2024-28/05/2024) |
| O        | System Implementation                 | N           | 2 (29/05/2024-30/05/2024) |
| P        | User Training                         | N,O         | 3 (31/05/2024-02/06/2024) |
| Q        | Data Migration                        | P           | 2 (03/06/2024-04/06/2024) |
| R        | User Manual                           | Q           | 3 (05/06/2024-07/06/2024) |
| S        | Technical Documentation               | R           | 3 (08/06/2024-10/06/2024) |
| T        | Release Notes                         | S           | 2 (11/06/2024-12/06/2024) |
| U        | Bug Fixes                             | S           | 3 (11/06/2024-13/06/2024) |
| V        | Updates and Upgrades                  | T,U         | 3 (14/06/2024-16/06/2024) |
| W        | User Support                          | U,V         | 5 (17/06/2024-21/06/2024) |

### 7.3 PERT CHART



7.4 GANTT CHART



## 8.0 BENEFIT OVERALL SUMMARY OF PROPOSED SYSTEM

Postgraduate students will find the Task Management System to be extremely beneficial as it facilitates the management of the project from inception to completion. The user may effortlessly locate a specific file by searching through the easily accessible folder that the system organizes the files into. Conversely, postgraduate projects are typically completed in groups, and the system offers postgraduate students a means of contact and collaboration. The project's files may be accessed by the other group members with permission from the group leader. The Task Management system facilitates communication and debate among the postgraduate students in the group. The system will notify the group members about the due dates, the to-do task and the message sent in the conversation space. Furthermore, while the project is ongoing, a security system guards the folder with a password and keeps it from leaking to the public.

To sum up, the Task Management System greatly aids users in finishing their projects. The research and publication processes are sped up and enhanced as the result of the system.

# PROJECT MANAGEMENT

## 1. URL of the GitHub Repository

<https://github.com/users/TANZHAOHONG/projects/4>

## 2. Repository Snapshot

